

SATHWIK DEVADIGA

Data Analyst | Market Risk & Risk Analytics

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PROFESSIONAL SUMMARY

Data Analyst with experience delivering Power BI dashboards, SQL/Python analytics and business-process automation, now specializing in Risk Management. Developed portfolio risk solutions covering VaR, Expected Shortfall, volatility, correlation, risk contribution, backtesting and hedging analysis. Targeting Market Risk / Risk Analyst roles where data analytics and automation can strengthen risk measurement, monitoring and reporting.

EXPERIENCE

Data Analyst — Hari Krishna Group (HK Group), Mumbai

Jun 2025 – Present

- **Develop and maintain** Power BI dashboards for B2B jewellery operations, Stock Turnover, Customer Snapshot, Weekly MTO, and Best Seller analysis to improve KPI visibility across business teams.
- **Analyze and transform** operational data using SQL, Excel, Power Query/DAX, and Python, streamlining data preparation and reducing manual processing effort by **40–50%**.
- **Automated** weekly invoice/PDF processing and email distribution workflows, reducing report-preparation time by **70–80%** and improving the reliability of scheduled deliverables.
- **Translate** stakeholder requirements into reusable data models and interactive dashboards, accelerating analysis turnaround by **25–35%** and supporting faster business reviews.
- **Assess inventory exposure** through stock ageing, turnover, sales velocity, and slow-moving inventory analysis, while evaluating **stock exposure to gold price movements** and their potential impact on inventory value and business risk.
- **Analyzing secondary sales trends**, customer and product performance, regional demand patterns, and changes in market conditions to identify sales movements and support business decision-making.

SDE Intern — Abans Group Pvt. Ltd.

Oct 2024 – Jan 2025

- Managed Excel-based operational datasets, produced lookups and charts, executed API testing with Postman, and supported a webhook-integrated WhatsApp chatbot, helping streamline recruitment and internal process workflows.

SELECTED RISK PROJECTS

Portfolio VaR & Expected Shortfall Engine

Python | Flask | Risk Analytics

- Engineered an interactive portfolio risk engine that calculates Historical, Parametric and Monte Carlo VaR, Expected Shortfall, annualized volatility and maximum drawdown from market-price data.
- Implemented correlation analysis, covariance-based risk contribution and rolling VaR backtesting; executed 10,000 Monte Carlo simulations per run to estimate portfolio loss under simulated return scenarios.
- Developed a configurable risk dashboard supporting custom tickers, portfolio weights, 95%/99% confidence levels and 1–10 day horizons, translating quantitative outputs into practical risk-monitoring views.

Portfolio Hedging & Options Risk

Python | Black-Scholes | Greeks

- Created a portfolio hedging module to estimate downside exposure and evaluate option-based protection using Black-Scholes pricing, Delta, Gamma, Vega and scenario analysis.
- Evaluated hedge effectiveness through cost-versus-protection and residual-risk analysis, shifting the project focus from standalone option pricing to portfolio risk management decisions.

Credit Risk Model — PD, LGD, EAD & Expected Loss

Python | Logistic Regression

- Structured borrower-level credit assessment around Probability of Default (PD), Loss Given Default (LGD), Exposure at Default (EAD) and Expected Loss ($EL = PD \times LGD \times EAD$).
- Developed an interpretable logistic-regression PD workflow with credit risk bands and model validation using ROC-AUC, KS statistic, confusion matrix and calibration, extending outputs to portfolio-level expected loss and exposure.

SKILLS

- **Risk & Finance:** VaR, Expected Shortfall, Portfolio Risk, Volatility, Correlation, Risk Contribution, Stress & Scenario Analysis, VaR Backtesting, Options & Greeks, Hedging, PD/LGD/EAD, Expected Loss
- **Programming & Data:** Python, Pandas, NumPy, SQL, Excel
- **BI & Visualization:** Power BI, DAX, Power Query, Matplotlib, Plotly
- **Quantitative Methods:** Statistics, Probability Distributions, Monte Carlo Simulation, Regression, Time Series, Financial Modelling
- **Tools & Development:** Flask, Git, GitHub

EDUCATION & CERTIFICATION

B.Sc. Information Technology — Jai Hind College, Mumbai University | CGPA: 8.9

May 2024

Introduction to Risk Management

Aug 2026

Programming, Data Structures and Algorithms Using Python — NPTEL-SWAYAM

Apr 2022